1.1 $\because J(w, w_0) = (y - Xw - w_0 \mathbf{1})^T (y - Xw - w_0 \mathbf{1}) + \lambda w^T w$

$\therefore \frac{dJ(w, w_0)}{dw_0} = (-2) \cdot \mathbf{1}^T (y - Xw - w_0 \mathbf{1})$

$\therefore \frac{dJ(w, w_0)}{dw_0} = 0 \Rightarrow \mathbf{1}^T (y - Xw - w_0 \mathbf{1}) = 0$

$\Rightarrow \mathbf{1}^T y - \mathbf{1}^T X w - \mathbf{1}^T w_0 \mathbf{1} = 0$

$\because \mathbf{1}^T y = \sum_{i=1}^{n} y_i$, $\mathbf{1}^T X = \sum_{i=1}^{n} x_i$, $\mathbf{1}^T w_0 = w_0 \cdot \mathbf{1}^T$

$\because X$ is centered $\therefore \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i = 0 \Rightarrow \mathbf{1}^T X = \sum_{i=1}^{n} x_i = n\bar{x} = 0$

$\therefore \Rightarrow \sum_{i=1}^{n} y_i - 0 - w_0 \mathbf{1}^T \mathbf{1} = 0$

$\Rightarrow \sum_{i=1}^{n} y_i - n w_0 = 0$

$\Rightarrow \hat{w}_0 = \frac{1}{n} \sum_{i=1}^{n} y_i = \bar{y}$

$\because \frac{dJ(w, w_0)}{dw} = -2X^T (y - Xw - w_0 \mathbf{1}) + 2\lambda w$

$\because \hat{w}_0 = \bar{y}$, $\frac{dJ(w, w_0)}{dw} = 0$

$\therefore -2X^T (y - Xw - \bar{y} \mathbf{1}) + 2\lambda w = 0$

$\therefore X^T (y - Xw - \bar{y} \mathbf{1}) - \lambda w = 0$

$\Rightarrow X^T y - X^T X w - \bar{y} X^T \mathbf{1} - \lambda w = 0$

$\because X$ is centered $\therefore X^T \mathbf{1} = 0$

$\therefore X^T y - X^T X w - \lambda w = 0$

$\therefore X^T y = X^T X w + \lambda w = (X^T X + \lambda I) w$

$\therefore w = (X^T X + \lambda I)^{-1} X^T y$

$\therefore$ The closed-form solution of cost function is

$$\begin{cases} \hat{w}_0 = \bar{y} \\ w = (X^T X + \lambda I)^{-1} X^T y \end{cases}$$

1.2 (i) $\because \phi(x_1) = [1, 0, 0]^T$

$\phi(x_2) = [1, \sqrt{2}, 1]^T$

$\therefore \phi(x_2) - \phi(x_1) = \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}$

$\therefore w$ is parallel to $\begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}$

(ii) $\because$ The margin $= \frac{1}{\|w\|}$

The margin is the distance from each support vector to the decision boundary.

The distance between $\phi(x_1)$ and $\phi x_2 = \|\phi(x_1) - \phi(x_1)\| = \sqrt{0+1+1} = \sqrt{2}$

$\therefore$ margin $= \frac{1}{2}\|\phi(x_2) - \phi(x_1)\| = \frac{\sqrt{2}}{2}$

(iii) $\because$ The margin $= \frac{1}{\|w\|} = \frac{\sqrt{2}}{2}$

$\therefore \|w\| = \frac{2}{\sqrt{2}} = \frac{2}{3}\sqrt{3}$

$\because$ According to (i), w is parallel to $\begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}$

$\therefore w = \begin{bmatrix} 0 \\ \frac{2\sqrt{2}}{3} \\ \frac{2}{3} \end{bmatrix}$

(iv) $\because$ The points will be on the decision boundary, and the constraints will be tight

$\therefore$ For $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \Rightarrow (-1)\left(\begin{bmatrix} 0 \\ \frac{2\sqrt{2}}{3} \\ \frac{2}{3} \end{bmatrix}^T \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + w_0\right) = 1 \Rightarrow w_0 = -1$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \left(\begin{bmatrix} 0 \\ \frac{2\sqrt{2}}{3} \\ \frac{2}{3} \end{bmatrix}^T \cdot \begin{bmatrix} \sqrt{2} \\ 1 \\ 1 \end{bmatrix} + w_0\right) = 1 \Rightarrow w_0 = -1$

$\therefore w_0 = -1$

(v) $f(x) = w_0 + w^T \phi(x)$

$= -1 + \begin{bmatrix} 0 & \frac{2\sqrt{2}}{3} & \frac{2}{3} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \sqrt{2}x \\ x^2 \end{bmatrix} = \frac{2}{3}x^2 + \frac{4}{3}x - 1$

2.1 Derivation of gradient in soft max logistic regression

The objective function of softmax regression:

$$J(w) = -\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{C} \left[\mathbb{I}(y_i = j) \log (f_{w_j b_j}(x_i)) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{C} \left[\mathbb{I}(y_i = j) \log \frac{\exp(w_j^T x + b_j)}{\sum_{c=1}^{C} \exp(w_c^T x + b_c)} \right]$$

For $z_j = w_j^T x_i + b_j$, $f_j = f_{w_j, b_j}(x_i) = \frac{\exp(z_j)}{\sum_{c=1}^{C} \exp(z_c)}$

$$\frac{dJ}{dz_k} = -\sum_{j=1}^{C} \left[\mathbb{I}(y_i = j) \frac{1}{f_j} \frac{df_j}{dz_k} \right]$$

$$= -\left(\sum_{j=k}^{C} \left[\mathbb{I}(y_i = j) \frac{1}{f_j} \frac{df_j}{dz_k} \right] + \sum_{j \neq k}^{C} \left[\mathbb{I}(y_i = j) \frac{1}{f_j} \frac{df_j}{dz_k} \right] \right)$$

$$= -\left(\left[\mathbb{I}(y_i = k) \frac{1}{f_k} \frac{df_k}{dz_k} \right] + \sum_{j \neq k}^{C} \left[\mathbb{I}(y_i = j) \frac{1}{f_j} \frac{df_j}{dz_k} \right] \right)$$

$$= -\left[\mathbb{I}(y_i = k) \cdot \frac{1}{f_k} \cdot \left(\frac{\exp(z_k)}{\sum_{c=1}^{C} \exp(z_c)} - \frac{\exp(z_k) \cdot \exp(z_k)}{\left(\sum_{c=1}^{C} \exp(z_c)\right)^2} \right) \right] - \sum_{j \neq k}^{C} \left[-\mathbb{I}(y_i = j) \frac{1}{f_j} \frac{\exp(z_j) \exp(z_k)}{\left(\sum_{c=1}^{C} \exp(z_c)\right)^2} \right]$$

$$= -\left(\mathbb{I}(y_i = k) \frac{1}{f_k} (f_k - f_k^2) - \sum_{j \neq k}^{C} \mathbb{I}(y_i = j) \frac{1}{f_j} f_j f_k \right)$$

$$= -\left(\mathbb{I}(y_i = k) - \sum_{j=1}^{C} \mathbb{I}(y_i = j) f_k \right) \quad \because \sum_{j=1}^{C} \mathbb{I}(y_i = j) = 1$$

$$\therefore = f_k - \mathbb{I}(y_i = k)$$

$$\therefore \frac{dJ}{dw_k} = \frac{1}{m} \sum_{i=1}^{m} \frac{dJ_i}{dz_k} \cdot \frac{dz_k}{dw_k} = \frac{1}{m} \sum_{i=1}^{m} (f_k - \mathbb{I}(y_i = k)) x_i$$